

Neoclassical Growth Model

$$Y_t = F(K_t, A_t L_t) \quad \text{production function}$$

$$I_t = K_{t+1} - (1 - \delta)K_t \quad \text{gross investment}$$

$$C_t + I_t \leq Y_t \quad \text{resource constraint}$$

$$L_t = (1 + n)^t L_0, \quad n \geq 0 \quad \text{labor = population}$$

$$A_t = (1 + g)^t A_0, \quad g \geq 0 \quad \text{technology (labor augmenting)}$$

$$\hat{U} = \sum_{t=0}^{\infty} \hat{\beta}^t u(\hat{c}_t) \frac{L_t}{H} \quad \text{rep. household's utility}$$

where H = number of households, $\frac{L_t}{H}$ = number of members of the rep. household, $\hat{c}_t = \frac{C_t}{L_t}$ = consumption per capita, and $\hat{\beta} \equiv \frac{1}{1+\rho}$ = discount factor, with $\rho > 0$ (ρ is the discount rate).

For any variable X , define:

$$x_t \equiv \frac{X_t}{A_t L_t}$$

By CRS: $\lambda Y_t = F(\lambda K_t, \lambda A_t L_t) \quad \forall \lambda > 0$. Choosing $\lambda = \frac{1}{A_t L_t}$: $\frac{Y_t}{A_t L_t} = F(\frac{K_t}{A_t L_t}, 1)$. Then:

$$y_t = f(k_t)$$

where $f(k_t) \equiv F(k_t, 1)$.

We can rewrite the resource constraint as follows:

$$C_t + K_{t+1} - (1 - \delta)K_t \leq Y_t$$

Dividing by $A_t L_t$:

$$\frac{C_t}{A_t L_t} + \frac{K_{t+1}}{A_{t+1} L_{t+1}} \frac{A_{t+1} L_{t+1}}{A_t L_t} - (1 - \delta) \frac{K_t}{A_t L_t} \leq \frac{Y_t}{A_t L_t}$$

Then:

$$c_t + k_{t+1}(1 + g)(1 + n) - (1 - \delta)k_t \leq f(k_t)$$

We can rewrite the utility function as follows:

$$\begin{aligned}
\hat{U} &= \sum_{t=0}^{\infty} \hat{\beta}^t u(\hat{c}_t) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \hat{\beta}^t u\left(\frac{C_t}{L_t}\right) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \hat{\beta}^t u\left(\frac{C_t A_t}{L_t A_t}\right) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \left(\frac{1}{1+\rho}\right)^t u(c_t A_t) \frac{L_0(1+n)^t}{H} \\
&= \frac{L_0}{H} \sum_{t=0}^{\infty} \left(\frac{1+n}{1+\rho}\right)^t u(c_t A_t) \\
&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t u(c_t A_t)
\end{aligned}$$

where $\beta \equiv \frac{1+n}{1+\rho}$. Assume $\rho > n$, so $\beta \in (0, 1)$.

**Particular Case: Log Utility, Cobb-Douglas
Technology, 100% Depreciation**

Assume:

$$u(\hat{c}_t) = \ln \hat{c}_t$$

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha} \quad \alpha \in (0, 1)$$

$$K_{t+1} = I_t \quad (i.e., \delta = 1)$$

Then:

$$\begin{aligned} y_t &\equiv \frac{Y_t}{A_t L_t} \\ &= \frac{K_t^\alpha (A_t L_t)^{1-\alpha}}{A_t L_t} \\ &= \frac{K_t^\alpha}{(A_t L_t)^\alpha} \\ &= \left(\frac{K_t}{A_t L_t} \right)^\alpha \\ &= k_t^\alpha \end{aligned}$$

Then:

$$c_t + (1 + g)(1 + n)k_{t+1} \leq k_t^\alpha$$

Also

$$\begin{aligned}\hat{U} &= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln(\hat{c}_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln(c_t A_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t (\ln A_t + \ln c_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln A_t + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t [\ln A_0 + t \ln(1 + g)] + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0}{H} \ln A_0 \sum_{t=0}^{\infty} \beta^t + \frac{L_0}{H} \ln(1 + g) \sum_{t=0}^{\infty} \beta^t t + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0 \ln A_0}{H} \frac{1}{1-\beta} + \frac{L_0 \ln(1+g)\beta}{H(1-\beta)^2} + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t\end{aligned}$$

Then:

$$\hat{U} = \text{constant}_1 + \text{constant}_2 \sum_{t=0}^{\infty} \beta^t \ln c_t$$

Since $\text{constant}_2 > 0$ we can work with:

$$U \equiv \sum_{t=0}^{\infty} \beta^t \ln c_t$$

The Planner's Problem

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln c_t$$

$$\text{s.t.} \quad c_t + (1+n)(1+g)k_{t+1} \leq k_t^\alpha$$

$$k_0 > 0 \text{ given}$$

Remark 1. It's clear that the constraint will be binding at the optimum.

Remark 2. We didn't include non-negativity constraints because our assumptions ensure that consumption and capital will be strictly positive at the optimum.

Bellman Equation

For any period t :

$$V(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

Solution Method: Value-function Iteration

1. Computation of V_1

Setting $V_0 \equiv 0$ we get:

$$V_1(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_{t+1} \geq 0$$

$$k_t \text{ given}$$

Solution:

$$k_{t+1} = 0$$

$$c_t = k_t^\alpha$$

$$V_1(k_t) = \alpha \ln k_t$$

Remark. We need to impose the constraint $k_{t+1} \geq 0$ explicitly, otherwise, with $V_0 \equiv 0$, the problem has no solution.

2. Computation of V_2

$$V_2(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V_1(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

From the first iteration we know $V_1(k_{t+1}) = \alpha \ln k_{t+1}$.
Then, using the constraint to eliminate k_{t+1} :

$$V_2(k_t) \equiv \max_{c_t} \ln c_t + \beta \alpha \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \alpha \beta \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \alpha \beta} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\alpha\beta} k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha \beta}{1 + \alpha \beta} k_t^\alpha$$

Then:

$$\begin{aligned} V_2(k_t) &= \ln\left(\frac{1}{1+\alpha\beta} k_t^\alpha\right) + \alpha\beta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta}{1+\alpha\beta} k_t^\alpha\right) \\ &= \ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha \ln k_t + \alpha\beta \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) + \beta\alpha^2 \ln k_t \\ &= \left[\ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha\beta \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) \right] + \alpha(1+\alpha\beta) \ln k_t \\ &= C + \alpha(1+\alpha\beta) \ln k_t \end{aligned}$$

3. Computation of V_3

$$V_3(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V_2(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

k_t given

Using our result for $V_2(k_{t+1})$:

$$V_3(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta[C + \alpha(1 + \alpha\beta) \ln k_{t+1}]$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

k_t given

Using the constraint to eliminate k_{t+1} :

$$V_3(k_t) \equiv \max_{c_t} \ln c_t + \beta C + \alpha\beta(1 + \alpha\beta) \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \alpha\beta(1 + \alpha\beta) \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\alpha\beta+(\alpha\beta)^2}k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2} k_t^\alpha$$

Then:

$$V_3(k_t) = \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}k_t^\alpha\right) + \beta \left[C + \alpha(1+\alpha\beta) \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta+(\alpha\beta)^2}{1+\alpha\beta+(\alpha\beta)^2} k_t^\alpha\right) \right]$$

$$V_3(k_t) = \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) + \alpha \ln k_t + \beta C + \alpha\beta(1+\alpha\beta) \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta+(\alpha\beta)^2}{1+\alpha\beta+(\alpha\beta)^2}\right) + \alpha\beta(1+\alpha\beta)\alpha \ln k_t$$

$$V_3(k_t) = \left[\ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) + \beta C + \alpha\beta(1+\alpha\beta) \ln\left(\frac{\alpha\beta(1+\alpha\beta)}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2]}\right) + \alpha[1+\alpha\beta+(\alpha\beta)^2] \ln k_t \right]$$

$$\begin{aligned}
V_3(k_t) = & \left[\ln \left(\frac{1}{1 + \alpha\beta + (\alpha\beta)^2} \right) + \beta \ln \left(\frac{1}{1 + \alpha\beta} \right) + \alpha\beta^2 \ln \left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)} \right) \right. \\
& + \alpha\beta(1 + \alpha\beta) \ln \left(\frac{\alpha\beta(1 + \alpha\beta)}{(1+n)(1+g)[1 + \alpha\beta + (\alpha\beta)^2]} \right) \Big] \\
& + \alpha[1 + \alpha\beta + (\alpha\beta)^2] \ln k_t
\end{aligned}$$

Then:

$$V_3(k_t) = D + \alpha[1 + \alpha\beta + (\alpha\beta)^2] \ln k_t$$

4. Computation of V_4

Using the same procedure, we get:

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3} k_t^\alpha$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha\beta[1 + \alpha\beta + (\alpha\beta)^2]}{1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3} k_t^\alpha$$

and

$$\begin{aligned}
V_4(k_t) = & \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3}\right) + \beta \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) \\
& + \beta^2 \ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha\beta^3 \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) \\
& + \alpha\beta^2(1+\alpha\beta) \ln\left(\frac{\alpha\beta(1+\alpha\beta)}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2]}\right) \\
& + \alpha\beta[1+\alpha\beta+(\alpha\beta)^2] \ln\left(\frac{\alpha\beta[1+\alpha\beta+(\alpha\beta)^2]}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3]}\right) \\
& + \alpha[1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3] \ln k_t
\end{aligned}$$

$$V_4(k_t) = E + \alpha[1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3] \ln k_t$$

5. Computation of V_j

From the previous steps we recognize the following pattern:

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}} k_t^\alpha$$

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} \frac{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2}}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}} k_t^\alpha$$

and

$$V_j(k_t) = e_j + d_j \ln k_t$$

where:

$$d_j = \alpha[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}]$$

$$e_j = a_j + b_j$$

$$a_j = \sum_{s=0}^{j-2} \beta^s \ln \left(\frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1-s}} \right)$$

$$b_j = \alpha\beta \sum_{s=0}^{j-2} \beta^s \left[\begin{array}{l} [1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2-s}] \\ \times \ln \left(\frac{\alpha\beta[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2-s}]}{(1+n)(1+g)[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1-s}]} \right) \end{array} \right]$$

6. Convergence

Letting $j \rightarrow \infty$ we obtain:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$$

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t$$

Notice that:

$$c_t = (1 - \alpha\beta) y_t$$

$$C_t = (1 - \alpha\beta) Y_t$$

Hence, the saving rate is constant:

$$\begin{aligned} s &= \alpha\beta \\ &= \frac{\alpha(1+n)}{1+\rho} \end{aligned}$$

Remark. The model looks like a Solow Model, except that now we know the determinants of the saving rate.

Solution Method: Guess and Verify

Guess:

$$V^G(k_t) = \Omega + \Theta \ln k_t$$

where Ω and Θ are parameters to be determined.

Now solve:

$$\tilde{V}(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V^G(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

The idea is to find $\tilde{V}(k_t)$ and then look for the values of Ω and Θ that make $\tilde{V}(k_t) = V^G(k_t) \forall k_t$.

We can rewrite the problem above as follows:

$$\tilde{V}(k_t) = \max_{c_t} \ln c_t + \beta \left[\Omega + \Theta \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right) \right]$$

FOC:

$$\frac{1}{c_t} + \beta\Theta \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \beta\Theta} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\beta\Theta} k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1 + \beta\Theta} k_t^\alpha$$

Then:

$$\tilde{V}(k_t) = \ln\left(\frac{1}{1+\beta\Theta} k_t^\alpha\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta} k_t^\alpha\right) \Rightarrow$$

$$\tilde{V}(k_t) = \left[\ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right) \right] + \alpha(1+\beta\Theta) \ln k_t$$

Then, $\tilde{V}(k_t) = V^G(k_t) \forall k_t$ requires:

$$\Omega = \ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right)$$

$$\Theta = \alpha(1+\beta\Theta)$$

From the second equation:

$$\Theta = \frac{\alpha}{1 - \alpha\beta}$$

Substit. into the first:

$$\Omega = \frac{1}{1 - \beta} \left[\ln(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \ln\left(\frac{\alpha\beta}{(1+n)(1+g)}\right) \right]$$

Then:

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln\left(\frac{\alpha\beta}{(1+n)(1+g)}\right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t$$

which coincides with the value function we found through value-function iteration.

Also, $c_t = \frac{1}{1+\beta\Theta} k_t^\alpha$ and $k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta} k_t^\alpha$ become:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$

and

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$$

which coincide with the expressions we found through value-function iteration.

Solution Method: FOC + B&S (Envelope) $\Rightarrow$ Euler

$$V(k_t) \equiv \max_{c_t, k_{t+1}} u(c_t) + \beta V(k_{t+1})$$

$$\text{s.t.} \quad c_t = f(k_t) - (1+n)(1+g)k_{t+1}$$

k_t given

Then:

$$V(k_t) \equiv \max_{k_{t+1}} u(f(k_t) - (1+n)(1+g)k_{t+1}) + \beta V(k_{t+1})$$

Remark. By using the constraint to eliminate c_t we have a problem where the state variable in period t is k_t and the control variable is k_{t+1} . The transition equation is simply $k_{t+1} = k_{t+1}$ (that is, the control variable in period t fully determines the value of the state in $t+1$). This is simpler than the alternative of using the constraint to replace k_{t+1} in $V(k_{t+1})$ because in the former case the transition equation for period t does not depend on the state at t , which makes the application of the envelope condition (the B&S formula) simpler (we discussed this during the first lecture).

FOC

$$-u'(f(k_t) - (1+n)(1+g)k_{t+1})(1+n)(1+g) + \beta V'(k_{t+1}) = 0$$

The expression above implicitly defines:

$$k_{t+1} = h(k_t)$$

Substituting $k_{t+1} = h(k_t)$ into the objective:

$$V(k_t) = u(f(k_t) - (1+n)(1+g)h(k_t)) + \beta V(h(k_t))$$

Differentiating:

$$\begin{aligned} V'(k_t) = & u'(f(k_t) - (1+n)(1+g)h(k_t)) [f'(k_t) - (1+n)(1+g)h'(k_t)] \\ & + \beta V'(h(k_t))h'(k_t) \end{aligned}$$

$$\begin{aligned} V'(k_t) = & u'(f(k_t) - (1+n)(1+g)h(k_t))f'(k_t) \\ & + \left[\begin{array}{l} -u'(f(k_t) - (1+n)(1+g)h(k_t))(1+n)(1+g) \\ + \beta V'(h(k_t)) \end{array} \right] h'(k_t) \end{aligned}$$

The second term in the r-h-s is zero by FOC. Then:

$$V'(k_t) = u'(f(k_t) - (1+n)(1+g)h(k_t)) f'(k_t)$$

Shifting the expression above one period ahead we get (B&S/Envelope Condition):

$$V'(k_{t+1}) = u'(f(k_{t+1}) - (1+n)(1+g)h(k_{t+1})) f'(k_{t+1})$$

Substituting into FOC, and using $h(k_{t+1}) = k_{t+2}$:

$$\begin{aligned} & -u'(f(k_t) - (1+n)(1+g)k_{t+1}) (1+n)(1+g) \\ & + \beta u'(f(k_{t+1}) - (1+n)(1+g)k_{t+2}) f'(k_{t+1}) = 0 \end{aligned}$$

This is the Euler equation. It's a second-order difference equation in k .

The two boundary conditions are:

$$k_0 = \bar{k}_0 \quad (\text{initial condition})$$

and

$$\lim_{t \rightarrow \infty} \beta^t u' (f(k_t) - (1 + n)(1 + g)k_{t+1}) k_{t+1} = 0 \quad (\text{TVC})$$

Remark 1. Read the note on the transversality condition (TVC).

Remark 2. We can write the Euler equation and the TVC as follows:

$$-u'(c_t)(1 + n)(1 + g) + \beta u'(c_{t+1}) f'(k_{t+1}) = 0$$

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0$$

For our particular case, the Euler Equation gives:

$$c_{t+1} = \frac{\alpha\beta}{(1 + n)(1 + g)} k_{t+1}^{\alpha-1} c_t$$

Remark 3. Check that our previous solution satisfies the Euler equation and the TVC.

Steady State

From $k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$ and $k_t = k_{t+1}$:

$$\begin{aligned} k_{ss} &= \left[\frac{\alpha\beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}} \\ &= \left[\frac{\alpha}{(1+\rho)(1+g)} \right]^{\frac{1}{1-\alpha}} \quad \text{since } \beta \equiv \frac{1+n}{1+\rho} \end{aligned}$$

From the policy function:

$$c_{ss} = (1 - \alpha\beta) k_{ss}^\alpha$$

From the production function:

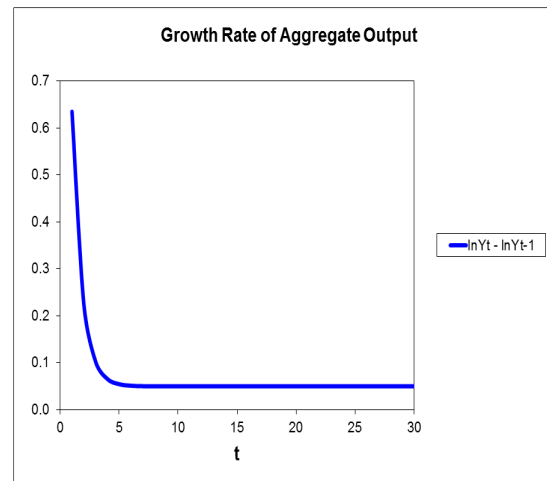
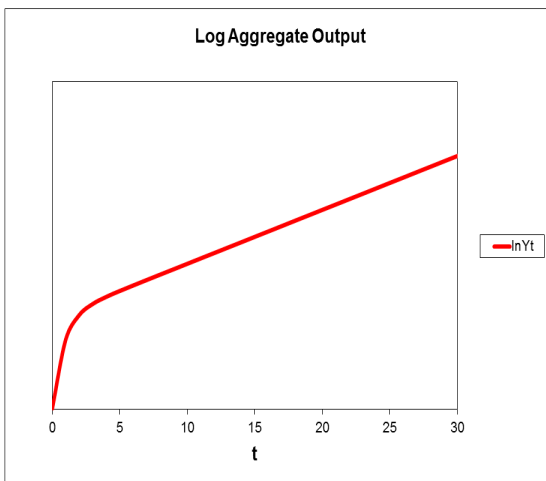
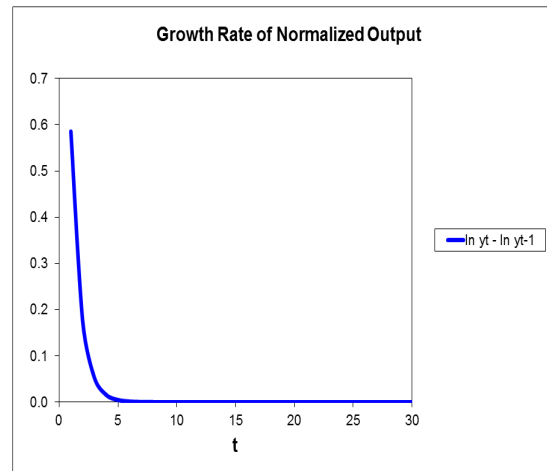
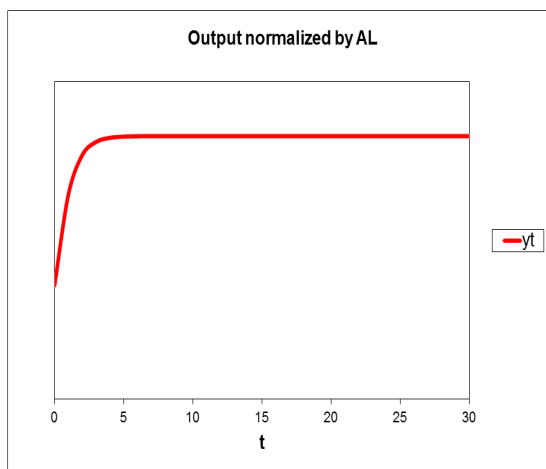
$$y_{ss} = k_{ss}^\alpha$$

From $k_t = \frac{K_t}{A_t L_t}$, $c_t = \frac{C_t}{A_t L_t}$, and $y_t = \frac{Y_t}{A_t L_t}$ we conclude that, in the s-s, the (gross) growth rate of K , C , and Y is $(1+g)(1+n)$ [balanced growth].

From $\hat{c}_t = \frac{C_t}{L_t}$ and $\hat{y}_t = \frac{Y_t}{L_t}$ we conclude that, in s-s, the (gross) growth rate of $\hat{c}_t$ and $\hat{y}_t$ is $1+g$.

Plots

Suppose: $\alpha = 0.3$, $\rho = 0.05$, $n = 0.03$, $g = 0.02$, $k_0 = 0.01$, $L_0 = 1$, $A_0 = 4$. These imply: $k_{ss} = 0.162$ and $y_{ss} = 0.58$.



Decentralized Version

Producer

$$\max_{K_t, L_t} K_t^\alpha (A_t L_t)^{1-\alpha} - w_t L_t - R_t K_t$$

FOC give:

$$R_t = \alpha k_t^{\alpha-1}$$

$$w_t = (1 - \alpha) A_t k_t^\alpha$$

Substituting these expressions into the objective function we see that profits are zero.

Household

Just for simplicity, normalize $H = 1$ so we have a unique household of size L_t at t .

$$\max_{\{C_t, I_t, K_{t+1}, B_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \hat{\beta}^t \ln \left(\frac{C_t}{L_t} \right) L_t$$

$$\text{s.t.} \quad C_t + I_t + B_t = w_t L_t + R_t K_t + (1 + r_{t-1}) B_{t-1}$$

$$I_t = K_{t+1} \quad (\text{since } \delta = 1)$$

$$K_0 > 0, B_{-1} = 0 \text{ given}$$

where we have imposed zero profits in the income side.

Since $L_t = L_0(1 + n)^t$ we can solve:

$$\max_{\{C_t, K_{t+1}, B_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln \left(\frac{C_t}{L_t} \right)$$

$$\text{s.t.} \quad C_t + K_{t+1} + B_t = w_t L_t + R_t K_t + (1 + r_{t-1}) B_{t-1}$$

$$K_0 > 0, B_{-1} = 0 \text{ given}$$

where $\beta \equiv \frac{1+n}{1+\rho} \in (0, 1)$.

The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \ln\left(\frac{C_t}{L_t}\right) + \sum_{t=0}^{\infty} \lambda_t [w_t L_t + R_t K_t + (1+r_{t-1})B_{t-1} - C_t - K_{t+1} - B_t]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow \beta^t \frac{1}{C_t} \frac{1}{L_t} = \lambda_t \Rightarrow \frac{\beta^t}{C_t} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \Rightarrow \lambda_t = \lambda_{t+1}(1 + r_t)$$

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = 0 \Rightarrow \lambda_t = \lambda_{t+1} R_{t+1}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow C_t + K_{t+1} + B_t = w_t L_t + R_t K_t + (1+r_{t-1})B_{t-1}$$

From the first two expressions we get: $\frac{\beta^t}{C_t} = \frac{\beta^{t+1}}{C_{t+1}}(1+r_t)$.
Then:

$$C_{t+1} = \beta(1 + r_t)C_t$$

From the second and third expressions we get:

$$R_{t+1} = 1 + r_t$$

Combining the two expressions above we get:

$$C_{t+1} = \beta R_{t+1} C_t$$

Dividing both sides by $A_{t+1}L_{t+1} = (1+g)A_t(1+n)L_t$ we get: $\frac{C_{t+1}}{A_{t+1}L_{t+1}} = \beta R_{t+1} \frac{C_t}{(1+g)A_t(1+n)L_t}$. Then:

$$c_{t+1} = \frac{\beta c_t}{(1+g)(1+n)} R_{t+1}$$

From the FOC for the firm we get: $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$. Substituting into the expression above we get:

$$c_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} k_{t+1}^{\alpha-1} c_t$$

which coincides with the Euler Equation we derived from the Planner's problem.

Market clearing in the credit market requires $B_t = 0 \forall t$ (closed economy). Imposing $B_t = 0 \forall t$ into the constraint we get: $C_t + K_{t+1} = w_t L_t + R_t K_t$. Since profits are zero, this becomes: $C_t + K_{t+1} = Y_t$. Dividing by $A_t L_t$ and rearranging appropriately we get:

$$c_t + (1 + g)(1 + n)k_{t+1} = k_t^\alpha$$

which coincides with the constraint faced by the Planner.

Hence, the equilibrium allocations of the decentralized competitive economy coincide with the ones chosen by the planner.

Stochastic Neoclassical Growth Model

(Brock and Mirman 1972)

Now assume:

$$Y_t = \theta_t F(K_t, A_t L_t)$$

where

$\{\theta_t\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables

By CRS:

$$y_t = \theta_t f(k_t)$$

Assume:

$$y_t = \theta_t k_t^{\alpha}$$

and

$$\ln \theta_t \sim \mathcal{N}(0, \sigma^2)$$

The planner chooses contingent plans for consumption and capital in order to solve:

$$\begin{aligned} \max \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln c_t \\ \text{s.t.} \quad & c_t + (1+n)(1+g)k_{t+1} = \theta_t k_t^\alpha \\ & k_0 > 0, \theta_0 > 0 \text{ given} \end{aligned}$$

where $\mathbb{E}_0$ is the conditional expectation as of time $t = 0$.

At time t the planner knows the realization of θ_t (and earlier) but not the future realizations. We define the state of the system as (k_t, θ_t) . Therefore, the BE is:

$$\begin{aligned} V(k_t, \theta_t) \equiv \max_{c_t, k_{t+1}} \quad & \ln c_t + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1}) \\ \text{s.t.} \quad & (1+n)(1+g)k_{t+1} = \theta_t k_t^\alpha - c_t \\ & k_t, \theta_t \text{ given} \end{aligned}$$

Using the constraint to eliminate k_{t+1} :

$$V(k_t, \theta_t) \equiv \max_{c_t} \ln c_t + \beta \mathbb{E}_t V \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)}, \theta_{t+1} \right)$$

Remark. Since c_t , k_t and θ_t are known at t , and the θ'_s are i.i.d., $\mathbb{E}_t$ runs over the distribution of θ_{t+1} .

We solve the problem using the Guess-Verify Method.

Guess:

$$V^G(k_t, \theta_t) = \Omega + \Theta \ln k_t + \Phi \ln \theta_t$$

Then:

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \mathbb{E}_t \left[\Omega + \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right) + \Phi \ln \theta_{t+1} \right]$$

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \Omega + \beta \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right) + \beta \Phi \mathbb{E}_t \ln \theta_{t+1}$$

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \Omega + \beta \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \beta \Theta \frac{1}{\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha$$

Then:

$$k_{t+1} = \frac{\theta_t k_t^\alpha - \frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta \Theta}{1 + \beta \Theta} \theta_t k_t^\alpha$$

Then:

$$\begin{aligned} \tilde{V}(k_t, \theta_t) = & \ln \left(\frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha \right) \\ & + \beta \mathbb{E}_t \left[\Omega + \Theta \ln \left(\frac{1}{(1+n)(1+g)} \frac{\beta \Theta}{1 + \beta \Theta} \theta_t k_t^\alpha \right) + \Phi \ln \theta_{t+1} \right] \end{aligned}$$

$$\begin{aligned}\tilde{V}(k_t, \theta_t) = & \left[\ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right) \right] \\ & + \alpha(1+\beta\Theta) \ln k_t + (1+\beta\Theta) \ln \theta_t\end{aligned}$$

Therefore, $\tilde{V}(k_t, \theta_t) = V^G(k_t, \theta_t) \forall (k_t, \theta_t)$ requires:

$$\Omega = \ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right)$$

$$\Theta = \alpha(1+\beta\Theta)$$

$$\Phi = 1 + \beta\Theta$$

From the second equation:

$$\Theta = \frac{\alpha}{1 - \alpha\beta}$$

Substituting into the third equation:

$$\Phi = \frac{1}{1 - \alpha\beta}$$

Substit. $\Theta = \frac{\alpha}{1-\alpha\beta}$ into the first equation:

$$\Omega = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right]$$

Then:

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t + \frac{1}{1-\alpha\beta} \ln \theta_t$$

and

$$c_t = (1-\alpha\beta)\theta_t k_t^\alpha$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha\beta\theta_t k_t^\alpha$$

Nonstochastic Steady State

Suppose $\ln \theta_t$ equals its mean in every period: $\ln \theta_t = 0 \forall t$. Hence: $\theta_t = 1 \forall t$.

Then, the evolution of the capital stock is:

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha \beta k_t^\alpha$$

Setting $k_{t+1} = k_t$ we get:

$$k_{nss} = \left[\frac{\alpha \beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}$$

where the subscript nss stands for "nonstochastic steady state".

Then:

$$c_{nss} = (1 - \alpha \beta) k_{nss}^\alpha$$

$$y_{nss} = k_{nss}^\alpha$$

Limiting (Ergodic) Distribution

From $k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha \beta \theta_t k_t^\alpha$ we get:

$$\begin{aligned} \ln k_{t+1} &= \ln \left(\frac{\alpha \beta}{(1+n)(1+g)} \right) + \alpha \ln k_t + \ln \theta_t \\ &= \psi + \alpha \ln k_t + \ln \theta_t \end{aligned}$$

Then:

$$\ln k_1 = \psi + \alpha \ln k_0 + \ln \theta_0$$

$$\begin{aligned} \ln k_2 &= \psi + \alpha \ln k_1 + \ln \theta_1 \\ &= \psi + \alpha(\psi + \alpha \ln k_0 + \ln \theta_0) + \ln \theta_1 \\ &= \psi(1 + \alpha) + \alpha^2 \ln k_0 + \alpha \ln \theta_0 + \ln \theta_1 \end{aligned}$$

$$\begin{aligned} \ln k_3 &= \psi + \alpha \ln k_2 + \ln \theta_2 \\ &= \psi + \alpha[\psi(1 + \alpha) + \alpha^2 \ln k_0 + \alpha \ln \theta_0 + \ln \theta_1] + \ln \theta_2 \\ &= \psi(1 + \alpha + \alpha^2) + \alpha^3 \ln k_0 + \alpha^2 \ln \theta_0 + \alpha \ln \theta_1 + \ln \theta_2 \end{aligned}$$

$$\begin{aligned}
\ln k_t &= \psi(1 + \alpha + \alpha^2 + \dots + \alpha^{t-1}) + \alpha^t \ln k_0 \\
&\quad + \alpha^{t-1} \ln \theta_0 + \alpha^{t-2} \ln \theta_1 + \dots + \alpha \ln \theta_{t-2} + \ln \theta_{t-1} \\
&= \psi \sum_{s=0}^{t-1} \alpha^s + \alpha^t \ln k_0 + \sum_{s=0}^{t-1} \alpha^s \ln \theta_{t-1-s} \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \sum_{s=0}^{t-1} \alpha^s \ln \theta_{t-1-s}
\end{aligned}$$

Notice that, given k_0 and $\ln \theta_0$, $\ln k_t$ is a linear combination of independent normally distributed random variables [$\ln \theta_t \sim \mathcal{N}(0, \sigma^2)$]. Therefore, given k_0 and $\ln \theta_0$, $\ln k_t$ is normally distributed. We can calculate its mean and variance. The mean is

$$\begin{aligned}
\mathbb{E}_0 \ln k_t &= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 \\
&\quad + \sum_{s=0}^{t-2} \alpha^s \mathbb{E}_0 \ln \theta_{t-1-s} \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 + \sum_{s=0}^{t-1} \alpha^s \times 0 \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0
\end{aligned}$$

The variance is:

$$\begin{aligned}
 Var_0\{\ln k_t\} &= Var_0\left\{\psi\frac{1-\alpha^t}{1-\alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 + \sum_{s=0}^{t-2} \alpha^s \ln \theta_{t-1-s}\right\} \\
 &= Var\left\{\sum_{s=0}^{t-2} \alpha^s \ln \theta_{t-1-s}\right\} \\
 &= \sum_{s=0}^{t-2} (\alpha^s)^2 Var\{\ln \theta_{t-1-s}\} \\
 &= \sum_{s=0}^{t-2} \alpha^{2s} \sigma^2 \\
 &= \sigma^2 \sum_{s=0}^{t-2} (\alpha^2)^s
 \end{aligned}$$

Then:

$$Var_0\{\ln k_t\} = \sigma^2 \frac{1 - (\alpha^2)^{t-1}}{1 - \alpha^2}$$

Taking limits for $t \rightarrow \infty$:

$$\begin{aligned}
\lim_{t \rightarrow \infty} \mathbb{E}_0 \ln k_t &= \lim_{t \rightarrow \infty} \left[\psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 \right] \\
&= \psi \lim_{t \rightarrow \infty} \frac{1 - \alpha^t}{1 - \alpha} + \ln k_0 \lim_{t \rightarrow \infty} \alpha^t \\
&\quad + \ln \theta_0 \lim_{t \rightarrow \infty} \alpha^{t-1} \\
&= \frac{\psi}{1 - \alpha} \quad [\alpha \in 0, 1)]
\end{aligned}$$

$$\begin{aligned}
\lim_{t \rightarrow \infty} Var_0 \{ \ln k_t \} &= \lim_{t \rightarrow \infty} \left[\sigma^2 \frac{1 - (\alpha^2)^{t-1}}{1 - \alpha^2} \right] \\
&= \frac{\sigma^2}{1 - \alpha^2} \lim_{t \rightarrow \infty} \left[1 - (\alpha^2)^{t-1} \right] \\
&= \frac{\sigma^2}{1 - \alpha^2}
\end{aligned}$$

From $y_t = \theta_t k_t^\alpha$:

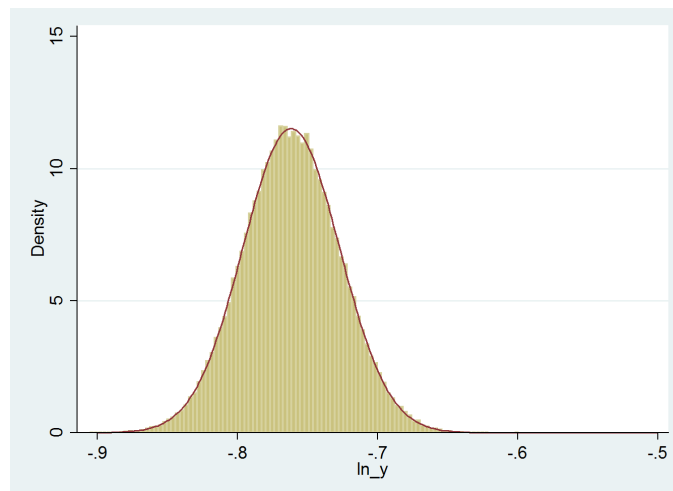
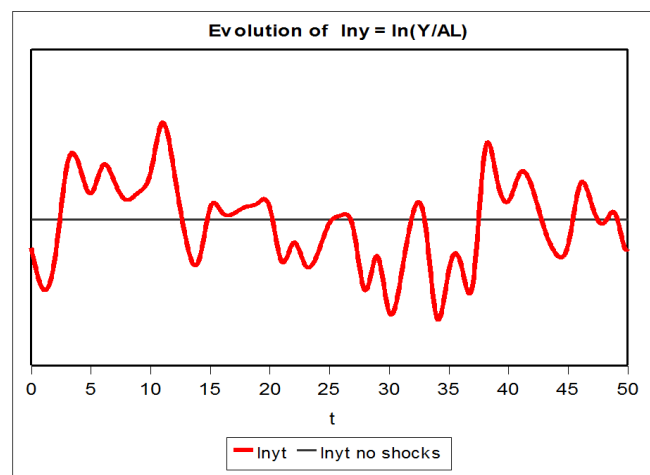
$$\ln y_t = \ln \theta_t + \alpha \ln k_t$$

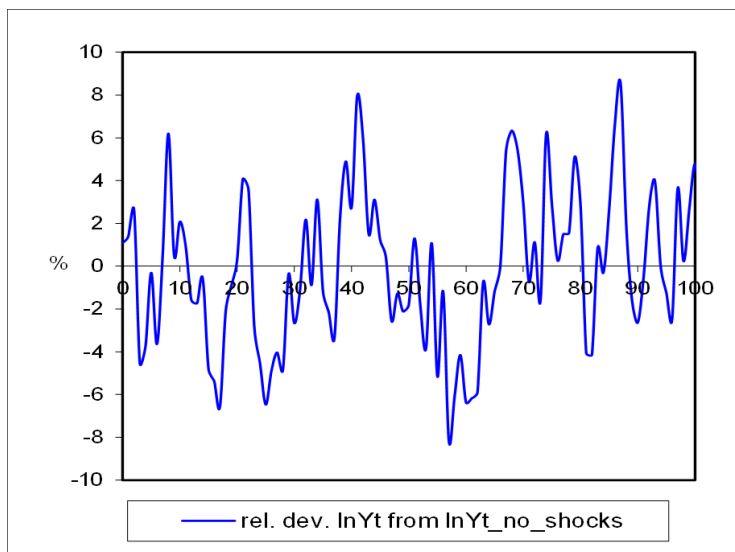
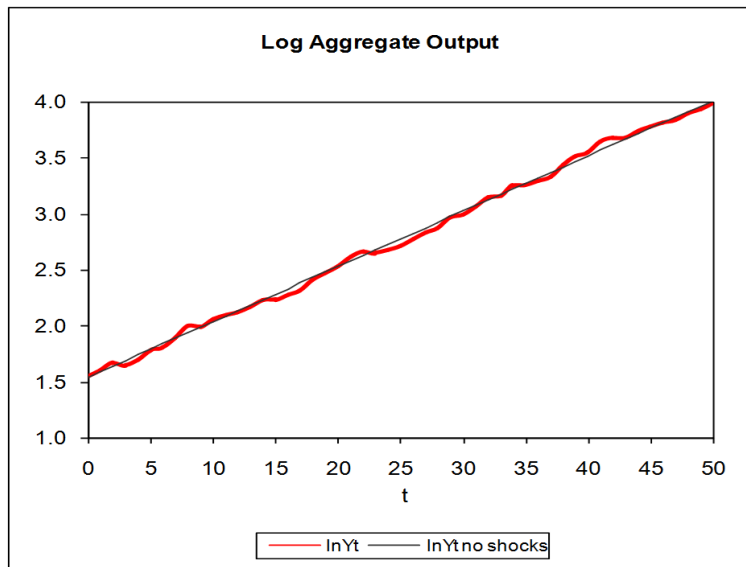
Hence, $\ln y_t$ is normally distributed, and we can calculate its mean and variance analogously.

Plots

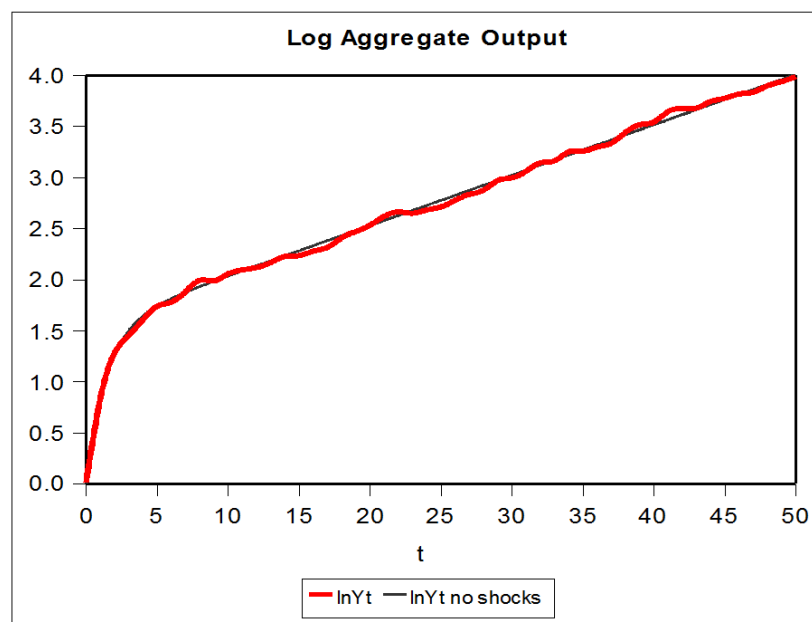
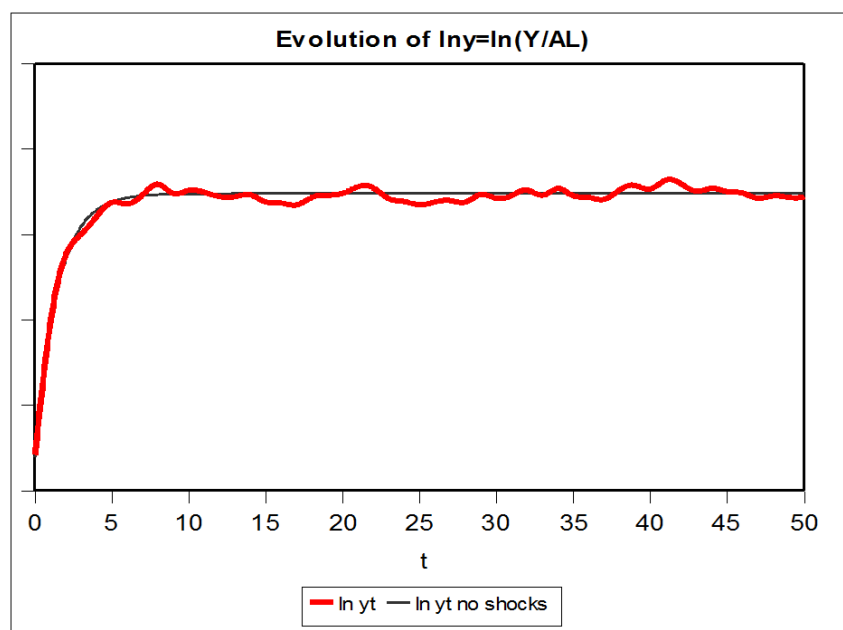
Assume $k_0 = k_{nss}$. Plot the evolution of $\ln y_t$ and $\ln Y_t$, and the empirical and limiting distributions of $\ln y_t$.

We assume $\alpha = 0.5$, $\rho = 0.05$, $n = 0.03$, and $g = 0.02$. These values imply: $k_{nss} = 0.218$ and $y_{nss} = 0.467$.



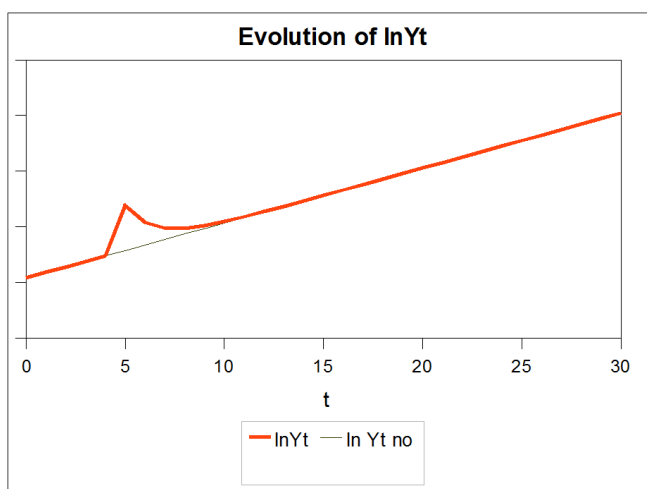
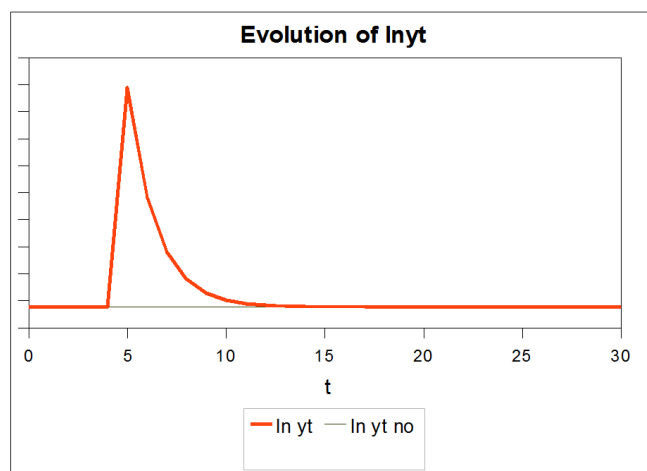


Now assume $k_0 < k_{nss}$. We get:



Impulse-Response Analysis

Set $\theta_t = 1$ ($\ln \theta_t = 0$) and assume the economy evolves along the balanced-growth path (nonstochastic steady state). At some period τ , θ_τ suddenly jumps up, and then returns to 1 from $\tau + 1$ onward.



Appendix

Phase Diagram for the Deterministic Model

From the Euler equation: $c_{t+1} - c_t = \left[\frac{\alpha\beta}{(1+n)(1+g)} k_{t+1}^{\alpha-1} - 1 \right] c_t$.

Using the resource constraint to eliminate k_{t+1} we get:

$$c_{t+1} - c_t = \left[\frac{\alpha\beta}{(1+n)(1+g)} \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)^{\alpha-1} - 1 \right] c_t.$$

The resource const. gives: $k_{t+1} - k_t = \frac{k_t^\alpha - c_t}{(1+n)(1+g)} - k_t$.

Hence, we have the following system of two difference equations in k and c :

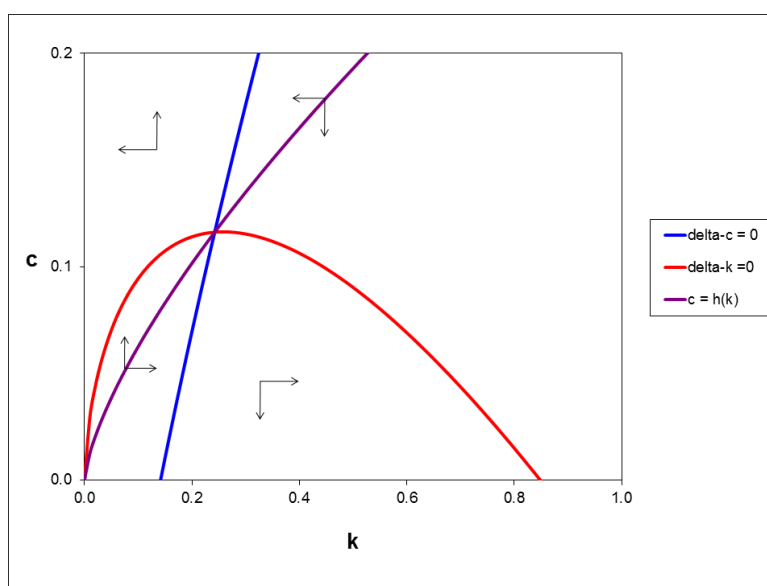
$$\begin{cases} \Delta c_{t+1} = \left[\frac{\alpha\beta}{(1+n)(1+g)} \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)^{-(1-\alpha)} - 1 \right] c_t \\ \Delta k_{t+1} = \frac{k_t^\alpha - c_t}{(1+n)(1+g)} - k_t \end{cases}$$

The $\Delta c_{t+1} = 0$ and $\Delta k_{t+1} = 0$ schedules are:

$$\left\{ \begin{array}{l} \Delta c_{t+1}=0: \quad c_t = k_t^\alpha - (1+n)(1+g) \underbrace{\left[\frac{\alpha\beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}}_{k_{ss}} \\ \Delta k_{t+1}=0: \quad c_t = k_t^\alpha - (1+n)(1+g)k_t \end{array} \right.$$

Now we plot the schedules given above together with the policy function:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$



Remark 1. For the plot we set: $\alpha = 0.7$, $\rho = 0.05$, $n = 0.03$, and $g = 0.02$. These values give: $k_{ss} = 0.242$, $y_{ss} = 0.371$, and $c_{ss} = 0.116$.

Remark 2. The three lines cross at $k = k_{ss}$.

Remark 3. The arrows indicate the qualitative dynamics of the system.

Remark 4. Notice that the policy function is the stable arm of the system.

Remark 5. In the continuous-time version of this model, $\Delta c_{t+1} = 0$ would produce a vertical line at $k = k_{ss}$.

Euler Equation and Transversality Condition**I. A Finite-Horizon Problem**

Consider the following problem:

$$\begin{aligned}
& \max \sum_{t=0}^T \beta^t u(c_t) \\
& \text{s.t. } c_t + k_{t+1} - (1 - \delta)k_t = f(k_t) \quad \forall t \in \{0, 1, \dots, T\} \\
& k_{T+1} \geq 0 \\
& k_0 > 0 \text{ given}
\end{aligned}$$

where:

$$u' > 0, u'' < 0, \lim_{c \rightarrow 0^+} u'(c) = \infty$$

$$f(0) = 0, f' > 0, f'' < 0, \lim_{k \rightarrow 0^+} f'(k) = \infty, \lim_{k \rightarrow \infty} f'(k) = 0$$

Remark. In principle, we should include a set of non-negativity constraints, $c_t \geq 0$ and $k_{t+1} \geq 0$. However, the Inada conditions $\lim_{c \rightarrow 0^+} u'(c) = \infty$ and $\lim_{k \rightarrow 0^+} f'(k) = \infty$ ensure that $c_t > 0$ and $k_{t+1} > 0$ at the optimum (except for k_{T+1} , that's why we kept the constraint $k_{T+1} \geq 0$). We should also write $c_t + k_{t+1} - (1 - \delta)k_t \leq f(k_t)$. But $u' > 0$ ensures this constraint will always hold with equality at the optimum.

The Lagrangian for the problem is:

$$\mathcal{L} = \sum_{t=0}^T \beta^t u(c_t) + \sum_{t=0}^T \lambda_t [f(k_t) + (1 - \delta)k_t - k_{t+1} - c_t] + \mu k_{T+1}$$

The Kuhn-Tucker first-order necessary conditions (FONC) are:

- (1) $\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \beta^t u'(c_t) = \lambda_t \quad \forall t \in \{0, 1, \dots, T\}$
- (2) $\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \Rightarrow \lambda_t = \lambda_{t+1} [f'(k_{t+1}) + (1 - \delta)] \quad \forall t \in \{0, 1, \dots, T-1\}$
- (3) $\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \lambda_T = \mu$
- (4) $\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} \quad \forall t \in \{0, 1, \dots, T\}$
- (5) $\frac{\partial \mathcal{L}}{\partial \mu} \geq 0 \Rightarrow k_{T+1} \geq 0$
- (6) $[CS] \quad \mu k_{T+1} = 0, \mu \geq 0$

where $[CS]$ is the *complementary-slackness* condition.

Euler Equation

Combining (1) and (2) we get: $\beta^t u'(c_t) = \beta^{t+1} u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)] \Rightarrow$

$$(7) \quad u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]$$

The interpretation of this expression is simple. Consider a (small) reduction of period- t consumption: $dc_t = -\Delta$, with $\Delta > 0$. This reduces utility by $u'(c_t)dc_t = -u'(c_t)\Delta$. The Δ units of capital accumulated through the reduction of current consumption increase future consumption by $dc_{t+1} = [f'(k_{t+1}) + (1 - \delta)]\Delta$. The increase in utility induced by this additional consumption is $\beta u'(c_{t+1})dc_{t+1} = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]\Delta$. The expression above says that, at the optimum, reshuffling consumption accross consecutive periods cannot increase utility: $dU = u'(c_t)dc_t + \beta u'(c_{t+1})dc_{t+1} = -u'(c_t)\Delta + \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]\Delta = [-u'(c_t) + \beta u'(c_{t+1})(f'(k_{t+1}) + (1 - \delta))]\Delta = 0$, where the last equality follows from (7).

Another way of looking at this is by rewriting the expression above as follows: $\frac{u'(c_t)}{\beta u'(c_{t+1})} = f'(k_{t+1}) + (1 - \delta)$. The left-hand side is the marginal rate of substitution between consumption in periods t and $t + 1$; the right-hand side is the marginal rate of transformation (*i.e.*, the rate at which current consumption can be transformed into future consumption through the available technology). At the optimum, the rate at which the consumer is *willing* to substitute consumption must be equal to the rate at which the consumer *can* substitute consumption.

Using (4) we can rewrite (7) as follows:

$$(8) \quad u'(f(k_t) + (1 - \delta)k_t - k_{t+1}) = \beta u'(f(k_{t+1}) + (1 - \delta)k_{t+1} - k_{t+2}) [f'(k_{t+1}) + (1 - \delta)]$$

This is the *Euler Equation*. It's a second-order difference equation in k . Solving this difference equation requires two boundary conditions. We already have one, the initial condition $k_0 > 0$. We'll use the complementary-slackness condition to get the second one.

Transversality Condition (TVC)

From (3) and (6) we get:

$$(9) \quad \lambda_T k_{T+1} = 0$$

From (1) we get:

$$(10) \quad \lambda_T = \beta^T u'(c_T)$$

Combining (9) and (10):

$$(11) \quad \beta^T u'(c_T) k_{T+1} = 0$$

Since $\beta > 0$ and $u'(c_T) > 0$, (11) implies:

$$(12) \quad k_{T+1} = 0$$

This is our second boundary condition, and is usually called the *transversality condition* (because it is a description of how the optimal path crosses the terminal line; to "transverse" means to "go across").

II. An Infinite-Horizon Problem

Now suppose the problem is:

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} - (1 - \delta)k_t = f(k_t) \quad \forall t \in \{0, 1, \dots\} \\ & k_0 > 0 \text{ given} \end{aligned}$$

The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [f(k_t) + (1 - \delta)k_t - k_{t+1} - c_t]$$

Then:

$$(13) \quad \frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \quad \beta^t u'(c_t) = \lambda_t \quad \forall t \in \{0, 1, \dots\}$$

$$(14) \quad \frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \Rightarrow \quad \lambda_t = \lambda_{t+1} [f'(k_{t+1}) + (1 - \delta)] \quad \forall t \in \{0, 1, \dots\}$$

$$(15) \quad \frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow \quad c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} \quad \forall t \in \{0, 1, \dots\}$$

Combining (13)-(15) we obtain the same Euler Equation as before:

$$(16) \quad u'(f(k_t) + (1 - \delta)k_t - k_{t+1}) = \beta u'(f(k_{t+1}) + (1 - \delta)k_{t+1} - k_{t+2}) [f'(k_{t+1}) + (1 - \delta)]$$

Again, we need two boundary conditions. We have one, the initial value of the capital stock, k_0 . But, how about the second? We don't have the complementary slackness condition to derive it (because there's no such thing as the "final" capital stock in an infinite-horizon setup). Fortunately, it is possible to derive a transversality condition by taking the limit for $T \rightarrow \infty$ of the TVC for the finite-horizon case. We have to be careful, however, because the relevant limit is not $\lim_{T \rightarrow \infty} k_{T+1}$. This doesn't make sense, since we've shown in (12)

that, for the finite-horizon case, $k_{T+1} = 0$. Instead, we have to go back one step and take the limit of expression (11), to get:

$$(17) \quad \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0$$

Now we can use (15) to write:

$$(18) \quad \lim_{T \rightarrow \infty} \beta^T u'(f(k_T) + (1 - \delta)k_T - k_{T+1}) k_{T+1} = 0$$

which is our TVC (and gives us the second boundary condition needed to solve the Euler Equation).

III. Some Remarks

Our informal argument for the derivation of the transversality condition for infinite-horizon models does not make clear whether the TVC is actually a *necessary* condition for an optimum. This has been debated quite a bit in the literature. There are papers showing conditions under which the TVC is necessary for an optimum. One of them is Kamihigashi (2002). For extensions to the stochastic case, see Kamihigashi (2003) and (2005). You can also check Karatzas and Sudderth (2010).

Stokey and Lucas (1989) [Theorem 4.15, page 98] give conditions under which the Euler Equation together with the TVC are *sufficient* conditions for an optimum.

It is common in the literature to eliminate c_t by substituting the constraint into the objective function. In this case we can define $F(k_t, k_{t+1}) \equiv u(f(k_t) + (1 - \delta)k_t - k_{t+1})$ and rewrite the TVC as follows:

$$(19) \quad \lim_{T \rightarrow \infty} \beta^T [-F_2(k_T, k_{T+1})] k_{T+1} = 0$$

From the TVC we know: $\lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0$. Equivalently, we can write: $\lim_{T \rightarrow \infty} \beta^{T-1} u'(c_{T-1}) k_T = 0$. But we know from the Euler Equation that: $u'(c_{T-1}) = \beta u'(c_T) [f'(k_T) + (1 - \delta)]$. Then, multiplying by k_T : $u'(c_{T-1}) k_T = \beta u'(c_T) [f'(k_T) + (1 - \delta)] k_T$. Substituting into the TVC: $\lim_{T \rightarrow \infty} \beta^{T-1} \beta u'(c_T) [f'(k_T) + (1 - \delta)] k_T = 0$. Then:

$$(20) \quad \lim_{T \rightarrow \infty} \beta^T u'(c_T) [f'(k_T) + (1 - \delta)] k_T = 0$$

which is an alternative form of the TVC. Notice, however, that this form of the TVC, unlike the original, was derived imposing the Euler Equation. If we define $F(k_t, k_{t+1}) \equiv u(f(k_t) + (1 - \delta)k_t - k_{t+1})$ as our objective function, (20) becomes:

$$(21) \quad \lim_{T \rightarrow \infty} \beta^T F_1(k_T, k_{T+1}) k_T = 0$$

which is the form of the TVC given in Stokey and Lucas (1989), page 98.

Sometimes the TVC is written using the Lagrange multiplier. From (9) we get:

$$(21) \quad \lim_{T \rightarrow \infty} \lambda_T k_{T+1} = 0$$

The TVC can be interpreted as saying that the present discounted value of wealth (the capital stock) at infinity must be zero; in other words, k_{T+1} should not grow too fast compared to its marginal value (shadow price) $\lambda_T = \beta^T u'(c_T)$. Hence, the TVC rules out overaccumulation of wealth. The idea is that if one saves too much and spends too little, then one is not behaving optimally.

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